A Soft Actor-Critic Method for Local Convergence Enhancement Under Strict Position–Orientation Constraints

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Abstract—Prior work has established the feasibility of deep reinforcement learning for random end-effector pose reaching. However, as the success tolerance is tightened from the centimeter scale to the millimeter scale, policy performance may deteriorate in the target neighborhood, with position and orientation constraints becoming difficult to satisfy simultaneously and deterministic pose-holding failures emerging. This paper studies how to improve the local convergence of Soft Actor-Critic (SAC) for high-precision position–orientation control without narrowing the target distribution or introducing an analytical controller. We introduce local Jacobian features, coupled pose-precision optimization signals, and precision-boundary experience replay to form an enhanced framework. The curriculum progressively tightens position and orientation tolerances, and multi-seed deterministic evaluation is used as the promotion criterion. Under a unified random target distribution, network architecture, precision curriculum, and fixed training budget, the proposed method achieves a success rate of 96.35% at a position–orientation tolerance of 10 millimeters and 0.10 radians, and further reaches a strict tolerance of 1 millimeter and 0.025 radians, where deterministic evaluation attains a success rate of 97.20%.

Index Terms—reinforcement learning, soft actor-critic, robotic manipulation, position–orientation control, curriculum learning, local convergence

I. Introduction

Random six-dimensional end-effector pose reaching requires a policy to handle three-dimensional position and three-dimensional orientation simultaneously. SAC improves exploration efficiency in continuous control through a maximum-entropy objective [1], while curriculum learning can organize training from easier tasks to progressively tighter target tolerances [2]. However, they do not guarantee corrective actions with appropriate direction and magnitude near the target, which is necessary for satisfying strict position–orientation constraints.

Existing studies have reported the feasibility of reinforcement learning for random pose control. For example, MAPPO-IK obtained millimeter-level position errors and small orientation errors on random position and orientation tasks [5]; continuous pose representations have also been combined with experience replay and curriculum learning to improve training convergence [6]. However, many random pose reaching studies mainly report final success rates or terminal errors, leaving the local performance degradation caused by continuously tightening success tolerances insufficiently analyzed.

This paper asks how local convergence degradation in high-precision position–orientation control can be reduced while keeping the target distribution, action space, and training budget fixed. We construct an enhanced framework around local kinematic guidance, coupled error optimization, and precision-region experience coverage, and analyze its performance through baseline comparison and component ablation.

The main contributions are as follows:

- We study performance degradation as SAC transitions from global reaching to fine convergence in the target neighborhood under strict position–orientation tolerances, and establish a fixed-budget, multi-stage precision curriculum with deterministic evaluation.
- Based on an existing keypoint representation of end-effector pose, we introduce Keypoint Jacobian local kinematic information, coupled pose-precision optimization signals, and precision-boundary experience resampling to construct an enhanced SAC framework.
- We evaluate local convergence at different precision stages through baseline comparisons and ablations under a unified training budget.

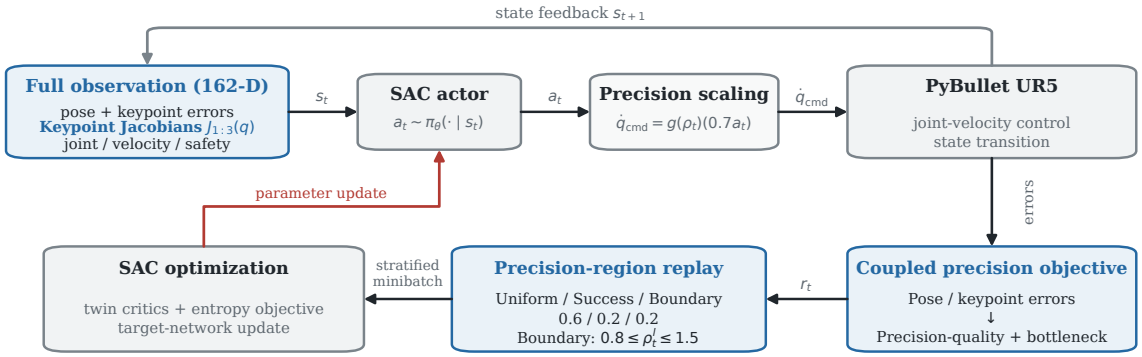


Fig. 1. Learning loop of Full. Keypoint Jacobians augment the actor observation, coupled pose signals form the reward, and uniform, success, and boundary subsets feed SAC optimization. Red arrows denote parameter updates.

II. Related Work

A. Continuous Control and Curriculum Learning

SAC balances exploration and return through a maximum-entropy objective [1]. Curriculum learning organizes tasks from easy to difficult and can improve the learnability of challenging tasks [2]; automated curricula for accurate motion planning likewise show the benefit of organizing training by task complexity [9]. In robotic manipulation, these mechanisms alleviate exploration over broad task ranges but do not directly ensure fine corrective actions near a pose target.

B. Pose Representation and Experience Replay

Pose representation affects error continuity and policy gradients. Continuous pose representations combined with replay and curriculum can improve convergence [6], while HER and PER alter experience reuse through goal relabeling or nonuniform sampling [3], [4]. Portela et al. encode pose with three end-effector-aligned keypoints [7]; observation–action alignment studies further motivate matching the representation to the control interface and reward structure [10]. We extend the keypoint representation with local Jacobian information for strict local convergence.

Multiple objectives in a complex reward can compete as position and orientation tolerances tighten. Reward curricula and nonlinear fusion have been studied to coordinate such objectives [11], [12]. Clustering replay and distribution-aware sampling likewise demonstrate the value of selective sampling for state coverage and distribution preservation [13], [14]. Our replay design specifically targets the coupled precision boundary rather than assigning priority from a generic transition score.

III. Problem Definition

Consider a six-degree-of-freedom robotic manipulator. At time t, the joint state is q_t, the policy outputs a joint-velocity action $a_t = \dot{q}_t$, the end-effector pose is $T(q_t) = [R(q_t), p(q_t)]$, and the target pose is $T_g = [R_g, p_g]$. The position and orientation errors are

$$e_p = \|p(q_t) - p_g\|_2, \qquad e_R = \|\log(R_g^{\mathsf{T}} R(q_t))\|_2. \tag{1}$$

Given position and orientation thresholds ϵ_p^l and ϵ_R^l, define the worst normalized constraint as

$$\rho_t^l = \max\left(\frac{e_{p,t}}{\epsilon_p^l}, \frac{e_{R,t}}{\epsilon_R^l}\right). \tag{2}$$

Let h_t denote the success-hold counter. When the position and orientation thresholds are continuously satisfied for $H = 5$ control steps, the success event at level l is defined as

$$\mathcal{S}_l = \mathbb{I}[\rho_t^l \le 1 \ \wedge \ h_t \ge H]. \tag{3}$$

The training objective is to maximize $\Pr(\mathcal{S}_l = 1)$ over the complete random target distribution.

IV. Method

A. Overall Framework and Base Representation

KP-SAC-Base combines explicit pose errors, the three-keypoint representation of Portela et al., keypoint-distance and progress rewards, and uniform replay. Under the same target distribution, network architecture, SAC settings, curriculum, and budget, Full adds Keypoint Jacobians, coupled precision signals, and precision-boundary replay (Fig. 1). M1, M2, and M3 remove bottleneck shaping, Boundary Replay, and Jacobian input, respectively, while preserving the remaining Full configuration.

B. Keypoints and Local Kinematic Input

Our keypoint construction follows the keypoint pose representation of Portela et al. [7]. A cube centered at the end-effector position and rotated with the end-effector orientation is constructed in the end-effector frame, and three vertices with direct correspondence are selected. Three non-collinear points with fixed correspondence jointly encode end-effector translation and rotation. Compared with directly concatenating Euler angles or quaternions, this

geometric representation converts the position–orientation error into point errors in a common coordinate space. For the fixed-base UR5 considered here, the cube has an edge length of 0.5 m; this is a virtual geometric scale for pose encoding rather than the physical tool size. The offsets of the three vertices in the tool frame are

$$u_1 = \tfrac{0.5}{2}[1,1,1]^\mathsf{T}, \qquad u_2 = \tfrac{0.5}{2}[1,-1,-1]^\mathsf{T},$$
$$u_3 = \tfrac{0.5}{2}[-1,1,-1]^\mathsf{T}.$$

The target and current keypoints are obtained by rotating these offsets with the corresponding pose and then translating them. Let $x_i(q)$ denote the selected current keypoint coordinates and x_i^g the target keypoints. The keypoint distance is

$$d_k = \frac{1}{K}\sum_{i=1}^{K} \|x_i(q) - x_i^g\|_2. \tag{4}$$

For each keypoint, the local velocity satisfies $\dot{x}_i = J_i(q)\dot{q}$, where $J_i \in \mathbb{R}^{3\times 6}$ is the keypoint Jacobian. Recent Jacobian-exploratory reinforcement learning work also uses local Jacobian features to augment the state representation [8]. Full concatenates normalized keypoint errors, keypoint progress, and local Jacobian information with the standard pose errors as the policy input. Specifically, keypoint errors are divided by 0.65 m and clipped to $[-1,1]$, while each J_i is divided by 1.1 and clipped to $[-1,1]$; the three J_i are concatenated into $9\times 6 = 54$ dimensions. The Base observation has 108 dimensions and the Full observation has 162 dimensions. To keep network capacity consistent, M3 retains the same input dimension and sets the corresponding 54 Jacobian features to zero. The observation also contains normalized joint positions and velocities, explicit position/orientation errors, end-effector velocity, and safety-related features. The Jacobian is computed from the UR5 kinematic model and the current joint state and is provided as a policy observation.

C. Joint Optimization Signals

With $\Delta e_p = e_{p,t-1} - e_{p,t}$ and $\Delta e_R = e_{R,t-1} - e_{R,t}$, the reward is summarized as

$$r_t = r_{\text{task}} + \lambda_p \Delta e_p + \lambda_R \Delta e_R + \lambda_k \Delta d_k - r_{\text{cost}}. \tag{5}$$

Here, r_{task} is the coupled success reward and r_{cost} covers action and collision costs. Base retains keypoint tracking, keypoint progress, continuous success rewards, and the action and collision costs. Full and M1–M3 additionally use level-normalized precision-quality and hold/leave terms. With $\rho_p = e_p/\epsilon_p^l$ and $\rho_R = e_R/\epsilon_R^l$,

$$q_{\text{pq}} = \exp\left[-\frac{\rho_p/0.05 + \rho_R/0.10}{2}\right], \tag{6}$$

where 0.05 and 0.10 are fixed empirical scales. Full further applies nonterminal bottleneck shaping to the worst normalized constraint:

$$\rho_b(s_t) = \max\left(\frac{e_{p,t}}{\epsilon_p^l}, \frac{e_{R,t}}{\epsilon_R^l}\right), \tag{7}$$

$$\Phi_b(s_t) = 5\exp[-\rho_b(s_t)/2], \tag{8}$$

$$r_{b,t} = (1-d_t)[\gamma\Phi_b(s_{t+1}) - \Phi_b(s_t)]. \tag{9}$$

Here, $d_t = 1$ denotes success, timeout, collision, or another hard-failure termination. Bottleneck shaping is zero on every terminal transition, while dedicated terms handle terminal outcomes; M1 removes only this potential term and retains precision quality and hold/leave. Table I lists the remaining definitions and fixed parameters.

TABLE I
Key training components of Full.

Component	Formulation / setting
Precision-quality shaping	q_{pq} as defined in (6); $w_{\text{prog}} = 2.0$
Precision-region retention	$r_{\text{hold}} = 0.15$, $r_{\text{leave}} = -0.2$
Success replay	$\mathcal{S}_l = 1$ transitions
Boundary replay	$0.8 \le \rho_t^l \le 1.5$
Replay mixture	Uniform / Success / Boundary = 0.6/0.2/0.2
Bottleneck shaping	$\Phi_b = 5\exp(-\rho_b/2)$

D. Precision-Boundary Experience Replay

Experience generated at loose curriculum stages does not sufficiently cover states near strict thresholds. For a transition (s_t, a_t, r_t, s_{t+1}), Full stores successful-episode transitions in the Success subset and uses the current-state condition $0.8 \le \rho_t^l \le 1.5$ for the Boundary subset; s_{t+1} distinguishes holding, leaving, and single-component failure. The fixed mixture increases sampling of state–action pairs near the coupled precision boundary without replacing the uniform replay stream. M2 removes only the Boundary component.

E. Automatic Precision Curriculum

Fig. 2 shows the shared curriculum. Let $\rho_t = \max(e_{p,t}/\epsilon_p, e_{R,t}/\epsilon_R)$. We use $g(\rho_t) = 0.05 + 0.05\rho_t$ for $\rho_t \le 1$ and $0.10 + 0.90\min(\rho_t - 1, 1)^2$ otherwise, with $\dot{q}_{\text{cmd}} = g(\rho_t)(0.7a_t)$. After at least 50,000 transitions per level, a deterministic probe runs every 300,000 transitions using three seeds and 300 episodes each. Promotion requires aggregate/seed success of 0.85/0.80, collision/timeout rates below 0.10/0.05, every task-space bin above 0.60, and a subsequent 1000-episode-per-seed confirmation under the same criterion. Online stochastic-policy success is retained only as a training diagnostic.

V. Experimental Setup

A. Environment and Training

IK is used only for offline target-reachability filtering and does not generate policy actions. Experiments are

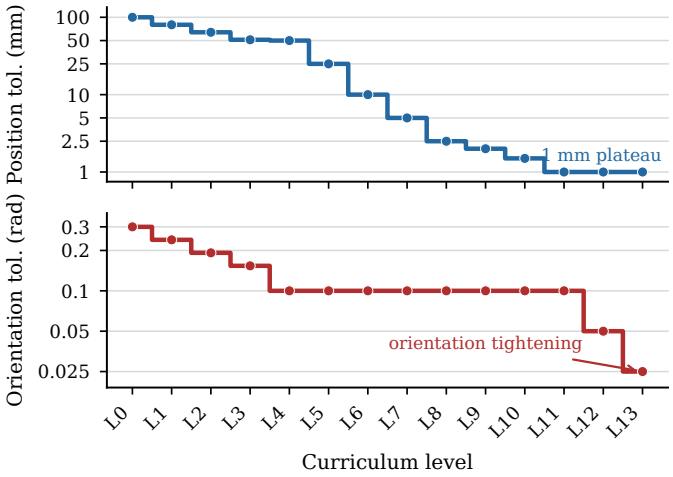


Fig. 2. Position and orientation tolerances across curriculum levels. Position tolerance reaches a 1-mm plateau at $L11$, after which orientation tolerance is tightened from 0.10 to 0.025 rad.

conducted in an obstacle-free PyBullet UR5 environment. Target positions are sampled from the initial end-effector position by first sampling $d \sim U(0.03, 0.50)$ m and a uniformly distributed direction on the unit sphere, followed by $p_g = p_0 + d\,v$. The target orientation difference is sampled as $\theta \sim U(0.03, \pi)$; the rotation axis is obtained by normalizing a three-dimensional standard Gaussian vector, and the axis-angle increment is composed with the initial orientation. Candidate targets are filtered for workspace validity, IK reachability, and self-collision. The filtered target distribution is shared across precision levels and methods. Each episode runs for at most 500 control steps, with a physics step of 0.00417 s and a control period of 0.05 s. Initial joint states are sampled within joint limits and filtered for self-collision. The policy action is a joint-velocity command. All methods use the same SAC hidden-layer structure, hyperparameters, target distribution, and budget of 6,000,000 transitions. Full and KP-SAC-Base use training seeds 11001 and 11002; M1, M2, and M3 use the prespecified single training seed 11001. Final frozen evaluation uses independent test seeds 31001, 31002, and 31003. Each policy checkpoint is evaluated deterministically for 1000 episodes per combination. Fig. 5 summarizes the common-tolerance comparison and highest reached precision. The single-seed ablations are used to compare performance after component removal and not to estimate variance across training seeds.

The component configurations of the comparison methods are listed in Table II. M3 preserves the 162-dimensional actor input by zeroing the 54 Jacobian channels.

B. Frozen-Policy Robustness Evaluation

Two frozen Full policies (training seeds 11001 and 11002) were tested at the strict 1 mm/0.025 rad tolerance, without retraining. Each policy was evaluated on 1000 episodes per condition, retaining the 500-step horizon and five-consecutive-sample success criterion. At each 20-Hz control cycle, position measurements received independent, zero-mean Gaussian translation noise with per-axis standard deviation σ_p; orientation measurements were perturbed separately by left-multiplicative SO(3) rotations with independent Gaussian rotation-vector components of standard deviation σ_R. The noisy pose consistently determined pose/keypoint features and precision-dependent action scaling. Joint states, velocities, and model-derived Jacobians remained exact. Success was determined from the true simulated state.

Mechanical disturbances were world-frame forces of fixed magnitude (0, 5, 10, or 20 N), applied at the current tool origin throughout each episode and reapplied at every 240-Hz physics substep. Force directions were sampled uniformly on the sphere every 0.25 s. Measurements remained exact in the force-only sweep. The combined sweep used 20 N with position noise and exact orientation measurements. Tasks and perturbation sequences were paired across conditions within each sweep, using random streams independent of task sampling. Single-perturbation evaluations used test-seed block 41001; the combined sweep used block 51001, separate from pilot tasks.

TABLE II
Ablation matrix. A check mark denotes an enabled component.

Method	KP	Jac.	Coupled	Boundary
Base	✓			
Full	✓	✓	✓	✓
M1	✓	✓		✓
M2	✓	✓	✓	
M3	✓		✓	✓

C. Evaluation Metrics

The main metrics are coupled pose success rate, timeout rate, collision rate, and the P95 terminal position and orientation errors. P95 is computed over all frozen-evaluation episodes, including both successful and failed episodes.

VI. Results and Analysis

A. Precision Milestones Under a Fixed Budget

The primary evaluation concerns curriculum progression under the fixed training budget. Fig. 4 reports when each precision milestone is first attained. Frozen evaluation covers the common L6 and each method's highest reached level, with 1000 episodes per method, test seed, and level. The common L6 supports direct comparison across all methods, whereas the highest reached level records fixed-budget progression. Full, M1, and M2 reach L13; M3 reaches L11; the two Base seeds reach L7 and L6. Results are not reported for levels a method does not reach, and all methods use the same promotion criterion.

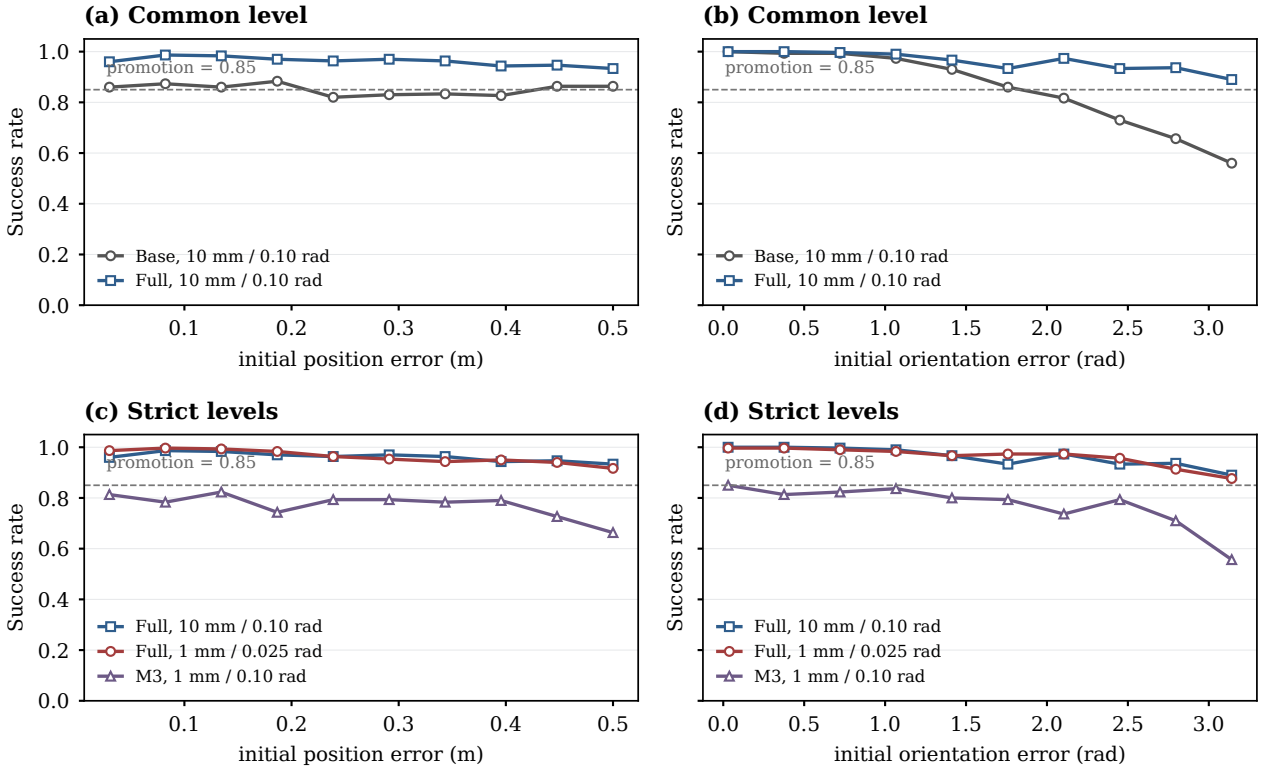


Fig. 3. Task-space success profiles under deterministic evaluation. Panels (a) and (b) show the common L6; panels (c) and (d) compare the stricter levels reached by Full and M3. The dashed line marks the 0.85 promotion criterion.

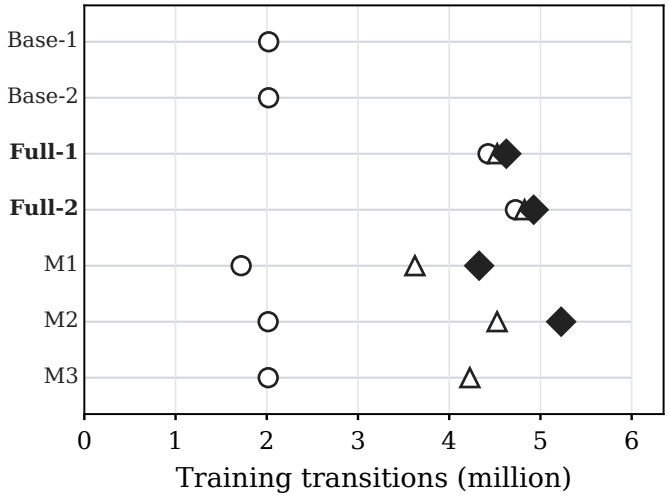


Fig. 4. Precision milestones within six million transitions. Circles, triangles, and diamonds denote first attainment of 10 mm/0.10 rad, 1 mm/0.10 rad, and 1 mm/0.025 rad.

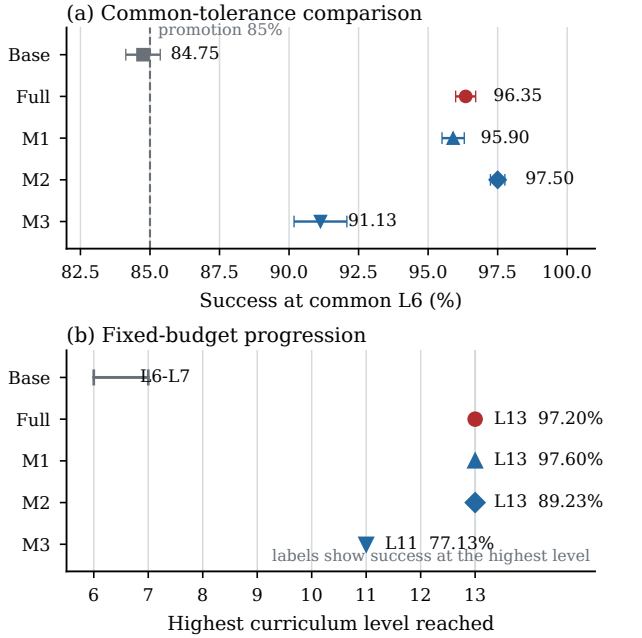


Fig. 5. Aggregate deterministic-evaluation results. (a) Success rate at the common 10 mm/0.10 rad tolerance; error bars show standard deviation. (b) Highest curriculum level reached within six million transitions; labels report success at that level. The Base range denotes the levels reached by its two training seeds.

B. Main Results

Fig. 5 combines the common-tolerance comparison with fixed-budget curriculum progression. For Full and Base, common-level statistics are computed over two training seeds and three test seeds; the ablations use one training seed and three test seeds. The key terminal P95 errors

(position in millimeters / orientation in radians) are: Base at 10 mm / 0.10 rad: $168.85 \pm 7.77/0.609 \pm 0.040$; Full at the same tolerance: $9.31 \pm 0.09/0.098 \pm 0.000$; Full at 1 mm / 0.025 rad: $0.098 \pm 0.000/0.024 \pm 0.000$; M2 at the same strict tolerance: $0.99 \pm 0.01/0.031 \pm 0.002$; and M3 at 1 mm / 0.10 rad: $0.351 \pm 0.056/0.097 \pm 0.001$. The terminal errors of M1 are close to those of Full.

At the common 10 mm / 0.10 rad tolerance, Full increases the success rate from 84.75% to 96.35% and reduces terminal position P95 from 168.85 mm to 9.31 mm. Full also reaches 1 mm / 0.025 rad within the fixed training budget, whereas the two Base seeds reach only 5 mm / 0.10 rad and 10 mm / 0.10 rad. Comparisons at stricter tolerances include only methods that reach the corresponding tolerance.

Fig. 3 further reports marginal success over the evaluation distance–orientation bins, exposing variation hidden by aggregate success rates.

At L6, Full remains between 93.3% and 98.7% across the initial-distance bins and between 89.0% and 100% across the orientation bins. The corresponding Base orientation profile decreases from 100% in the smallest bin to 56.0% in the largest bin. At the strict levels, Full-L13 remains above the 0.85 promotion line in every marginal bin, whereas M3-L11 falls to 66.3% in the largest distance bin and 55.7% in the largest orientation bin. These profiles localize the aggregate degradation toward more difficult initial orientations and, for M3, larger initial displacements.

C. Ablation Results

M1, M2, and M3 achieve 95.90%, 97.50%, and 91.13% at 10 mm / 0.10 rad, respectively. M3 has a larger terminal position-error P95 and progresses only to 1 mm / 0.10 rad. Full achieves 97.20% at 1 mm / 0.025 rad, compared with 89.23% for M2; M2 has a position-error P95 of 0.99 mm and an orientation-error P95 of 0.031 rad, so the degradation is concentrated in the orientation-error tail. M1 achieves 97.60% at the same strict tolerance, close to Full's 97.20%, and no clear independent gain from Bottleneck shaping is observed in the current experiment.

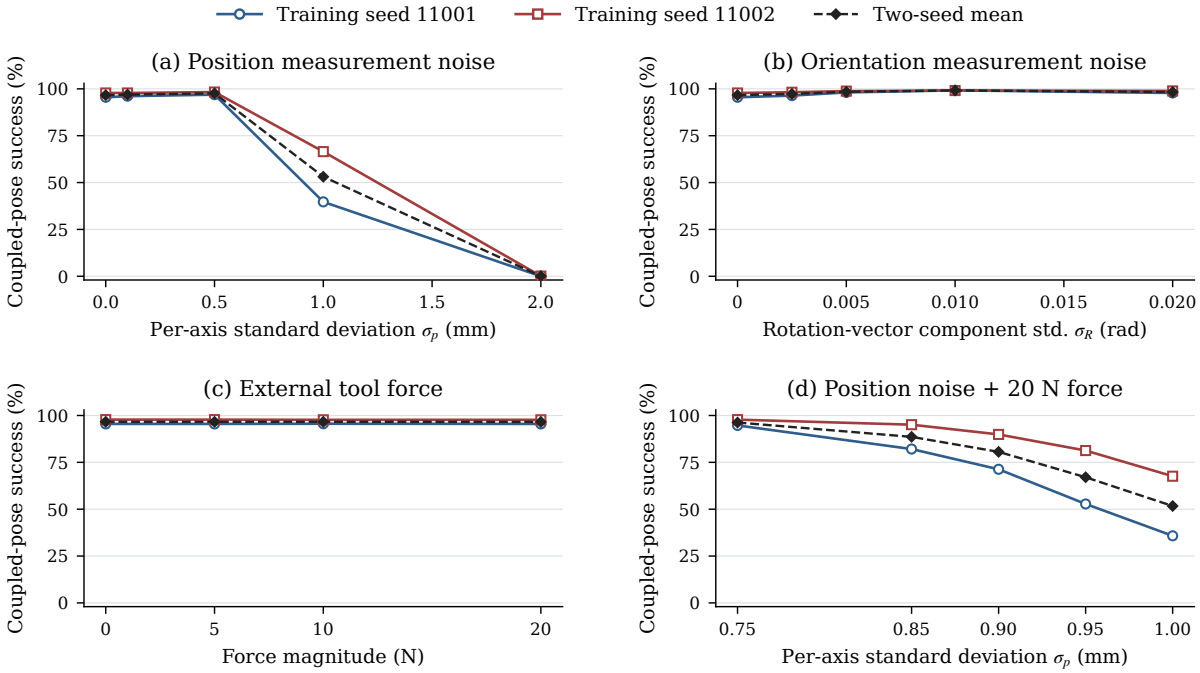


Fig. 6. Frozen Full policies at 1 mm/0.025 rad under (a) position measurement noise, (b) orientation measurement noise, (c) randomly redirected external tool force, and (d) position noise with a 20-N tool force. Colored curves show two training seeds; the dashed curve is their mean. Each policy uses 1000 episodes per condition. Noise magnitudes are standard deviations per translation or rotation-vector component. Panels (a) and (b) share the zero-noise evaluation. Tasks are paired within each sweep; panel (d) uses a different task block. Success uses the true simulated state.

D. Sensitivity to Measurement Noise and External Forces

Fig. 6 summarizes the two frozen policies and their arithmetic mean. Panel (d) displays the combined position-noise sweep over $\sigma_p = 0.75$–1.00 mm; this range was restricted after evaluation, with the two higher evaluated levels retained in the experiment archive. Without perturbations, mean success was 96.65% on this test set. With position noise, success was 97.60% at $\sigma_p = 0.5$ mm, but fell to 53.10% at 1 mm and 0.05% at 2 mm. Thus, noise on the scale of the position-success tolerance substantially degraded reaching performance. Under the tested nonzero orientation-noise levels (0.0025–0.020 rad per component), mean success ranged from 97.30% to 99.15%. Small nonmonotonic increases do not establish

a general benefit from measurement noise.

Force-only mean success remained 96.65% at 0, 5, and 10 N and 96.60% at 20 N. This characterizes the policy together with its underlying velocity servo; it does not isolate disturbance rejection by the learned policy. With simultaneous 20-N force and position noise, mean success decreased from 96.25% at 0.75 mm to 88.60%, 80.60%, 67.05%, and 51.70% at 0.85, 0.90, 0.95, and 1.00 mm, respectively. At 1.00 mm, the individual policies achieved 35.8% and 67.6%, exposing sensitivity to the training seed.

VII. Discussion and Limitations

The conclusions are limited to the obstacle-free PyBullet simulation and the specified random target distribution. Additional frozen-policy tests model independent Gaussian end-effector measurement noise and randomly redirected external forces, but do not cover encoder noise, gearbox backlash, calibration bias, communication delay, or real-robot dynamics. Force robustness concerns the complete policy–velocity-servo system. Thus, 1 mm denotes a simulated position-success tolerance, not demonstrated physical-robot accuracy. Future work should address these unmodeled effects and physical validation.

M1–M3 are targeted component ablations and use a single training seed; they therefore provide evidence of performance changes associated with component removal, not estimates of cross-seed variance or complete module-level causality. In particular, the M3 degradation is consistent with removing Jacobian information under fixed network capacity, but the evaluation does not directly measure action-direction quality or exclude other forms of local directional guidance. Similarly, M2 associates Boundary Replay removal with a larger orientation-error tail, while M1 does not establish an independent benefit from bottleneck shaping in the current setting.

VIII. Conclusion

This paper presented an SAC framework for local convergence enhancement under strict position–orientation constraints. The proposed framework raises success at 10 mm/0.10 rad from KP-SAC-Base's 84.75% to 96.35% and reaches 1 mm/0.025 rad. Without Boundary Replay, M2 achieves 89.23% at this strict tolerance; without Jacobian features, M3 reaches only 1 mm/0.10 rad with 77.13% success. No clear independent gain from Bottleneck shaping is observed.

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